

Design of a LNA with BFP460 transistor

Gabriel Venegas, Santiago Cadena

gvenegas, scadena @unal.edu.co

Departamento de Ingeniería Eléctrica y Electrónica
Universidad Nacional de Colombia. Bogotá D.C., Colombia.

Abstract—This paper presents the design, simulation, and analysis of a Low Noise Amplifier (LNA) based on the BFP460 transistor. Key parameters such as impedance matching (S_{11}), power gain, noise figure (NF), power consumption, and 1 dB compression point (P_{1dB}) are evaluated. Results were obtained through simulations in LTspice, demonstrating compliance with RF specification. The specifications required for designing the LNA for accomplish with criteria are:

- Select the frequency range for performance of the LNA.
- Determine the gain to amplify weak signals.
- Input/Output Impedance: Define input and output impedance to match with other componentes.
- Set limits on power consumption to ensure efficiency.

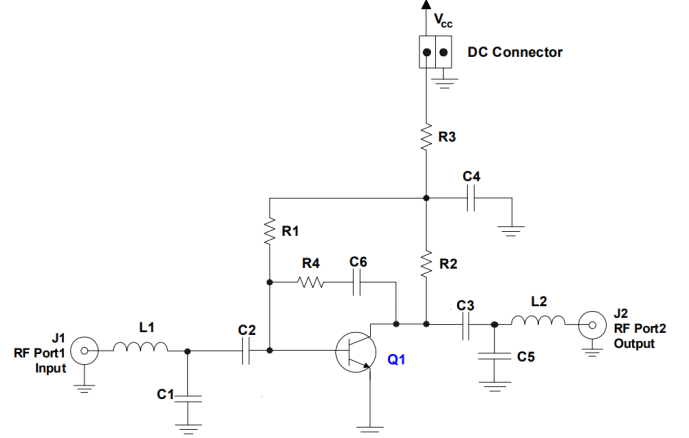


Figure 1: LNA Schematic

I. INTRODUCTION

A. Goals

- Design an LNA with high gain and low noise.
- Optimize impedance matching to 50Ω .
- Analyze performance using S, NF and P_{1dB} parameters.

B. Design specifications

TABLE I: LNA Specifications

Parameter	Value
S_{11}	$< -10dB$
Operation Frequency	$87.5108MHz$
Objective Gain	$> 15dB$
Noise Figure (NF)	$< 3dB$
Power Consumption	best possible
P_{1dB}	$> -10dB$

II. CIRCUIT ANALYSIS AND JUSTIFICATION FOR DESIGN

A. Chosen Topology

The selected Low Noise Amplifier (LNA) topology is a common-emitter silicon bipolar junction transistor (BJT) amplifier. This configuration is widely used in RF applications due to its ability to amplify weak signals with a good trade-off between gain, low noise figure, and stability.

The topology includes:

- Inductive matching at the input and output for 50Ω impedance adaptation.
- A resistive bias network that sets a stable operating point.
- Negative feedback through R4 and C6 to enhance stability and linearity.

B. Transistor Selection: BFP460

The Infineon BFP460 transistor was chosen due to its suitability for RF applications:

- High gain performance (~ 20 dB in the FM band).
- Low noise figure ($NF < 2$ dB).
- High transition frequency ($f_t > 5$ GHz), ensuring good performance in the 88–108 MHz FM band.
- Suitable for surface-mount designs (SOT343 package).

The 2.6 V supply variant was chosen over the 1.8 V version due to its wider dynamic range, making it more robust for active FM antennas or preamplifiers.

C. Component Selection and Justification (2.6 V configuration)

TABLE II: Component Justification for the 2.6 V BFP460 LNA Design

Component	Justification
C1 (12 pF)	Input matching capacitor. Forms a resonant network with L1 for 50 Ω impedance matching.
C2, C3, C6 (330 pF)	DC blocking capacitors that allow RF signal transmission while isolating DC levels between stages.
C4 (47 nF)	RF decoupling capacitor for the DC power supply. Shorts RF signals to ground, avoiding unwanted coupling through V_{CC} .
C5 (12 pF)	Output matching component. Used along with L2 to provide proper output impedance matching.
R1 (47 k Ω)	Sets the base bias current for the transistor. High enough to avoid affecting the input impedance.
R2 (68 Ω)	Biasing resistor that also aids in output matching and contributes to gain control and stability.
R3 (68 Ω)	Works with R2 to improve DC biasing and RF decoupling; part of the negative feedback loop.
R4 (1.8 k Ω)	Provides negative feedback, which reduces gain variation and improves stability and linearity.
L1 (100 nH)	Input matching inductor. Resonates with C1 and the input impedance of the transistor.
L2 (39 nH)	Output matching inductor. Works with C5 to ensure maximum power transfer to a 50 Ω load.

D. Impedance Matching to 50 Ω

This design ensures input and output impedance matching to 50 Ω , which is essential to minimize reflections and maximize power transfer between the antenna, the LNA, and subsequent RF stages.

E. Suitability for FM Band (88–108 MHz)

All selected inductors and capacitors are dimensioned to provide resonant behavior or act as RF short/open circuits within the FM band. The values were chosen based on their proven effectiveness in real-world applications for achieving high gain, stability, and impedance matching within the 88–108 MHz range.

F. Bias Point Selection: Determining V_{CE} and I_C

To ensure proper linear amplification and thermal stability, the bias point was selected based on typical recommendations for the BFP460 operating at 2.6 V.

G. Power Supply and Bias Current Justification

The supply voltage was set to $V_{CC} = 2.6$ V and the collector current to $I_{CC} = 4.6$ mA based on the recommended typical operating point for the BFP460 transistor in LNA applications. These values were selected to optimize the trade-off between gain, noise figure (NF), and linearity while maintaining low power consumption.

According to the Infineon datasheet and application notes, the BFP460 achieves a low noise figure and high gain at this collector current when biased at 2.6 V. The chosen bias point ensures that the transistor operates in the active region with sufficient headroom for linear amplification of RF signals in the FM band (88–108 MHz).

The current I_{CC} was calculated using Ohm's law and the voltage drop across the emitter resistor network (R2 and R3), which also provides thermal stability:

$$I_{CC} \approx \frac{V_{CC} - V_{CE}}{R2 + R3}$$

Assuming a collector-emitter voltage drop of approximately 1.3 V (typical for this class of amplifier), and using the resistor values:

$$I_{CC} = \frac{2.6 \text{ V} - 1.97 \text{ V}}{68 \Omega + 68 \Omega} \approx 4.6 \text{ mA}$$

This current is sufficient to bias the transistor for optimal low-noise amplification while avoiding excessive power dissipation, which is important for thermally constrained RF front ends such as those in active FM antennas.

III. SIMULATIONS AND ANALYSIS

The chosen topology was now implemented in LTSpice in order to simulate the LNA based on the spice model provided by Infineon for the RF transistor BFP460. The schematic for the simulation is presented next.

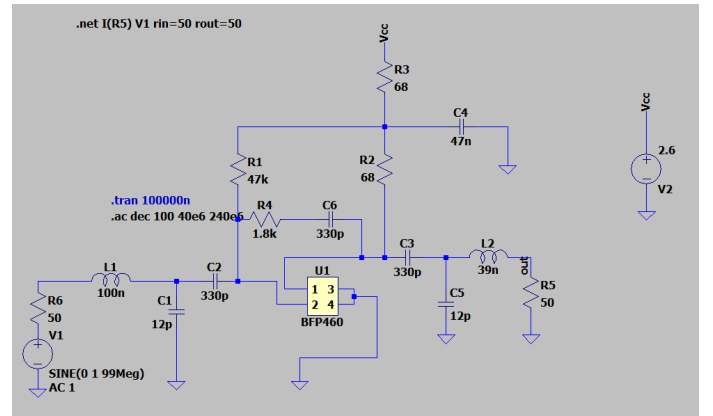


Figure 2: LNA Schematic in LTSpice

A. S11 Parameter (Impedance Matching)

In first place the S11 parameter was evaluated as shown next.

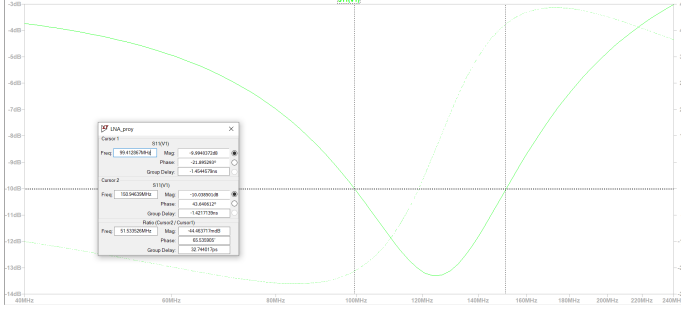


Figure 3: S11 plot in dB. $S_{11} < -10\text{dB}$ is met in the band of interest.

Viewing the S11 parameter of an LNA (Low-Noise Amplifier) helps assess its input impedance matching. A low S11 (or high return loss) indicates good matching, minimizing signal reflections and ensuring maximum power transfer from the source to the amplifier. In this case the S11 parameter obtained for the FM frequencies is under the -10 dB, meeting the required specification.

B. Power Gain

$$G_{\text{dB}} = 10 \log_{10} \left(\frac{P_{\text{out}}}{P_{\text{in}}} \right) \quad (1)$$

The Gain (dB) of an LNA indicates how much it amplifies the input signal. A higher gain means better signal amplification, which is crucial for improving weak signals in RF systems.

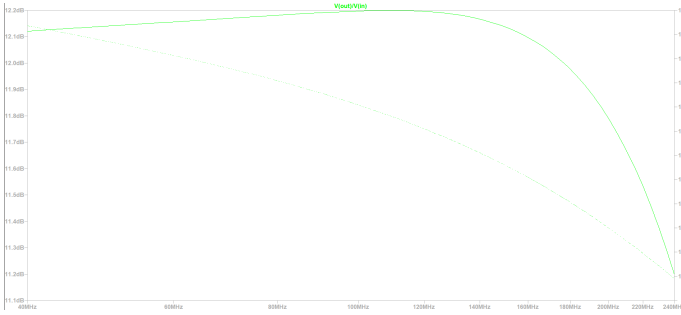


Figure 4: LNA Gain as a function of frequency.

The simulation shows that over the entire range of 40 MHz to 150 MHz the gain provided by the LNA is greater than 12 dB.

C. Noise Figure (NF)

The Noise Figure (NF) of an LNA measures how much it degrades the signal-to-noise ratio (SNR). A lower NF (closer to 0 dB) means the amplifier adds minimal noise, which is critical for maintaining sensitivity in weak-signal applications like radio astronomy or communications. Ideally, an LNA should have both low NF and sufficient gain to preserve signal quality.

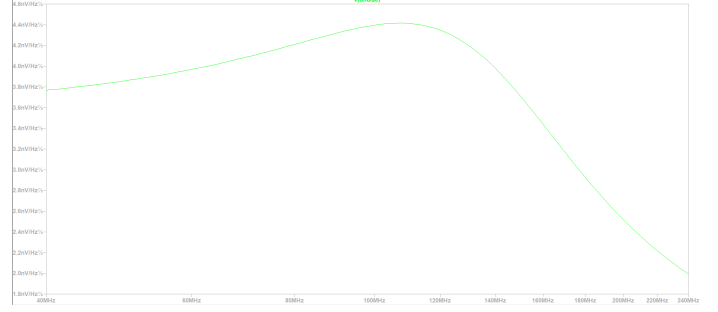


Figure 5: Simulated Noise Figure.

IV. DEVICE IMPLEMENTATION AND ANALYSIS

A. PCB Design

Below are the schematics resulting from the PCB design in Kicad. For the design, SMD components were put on a single layer.

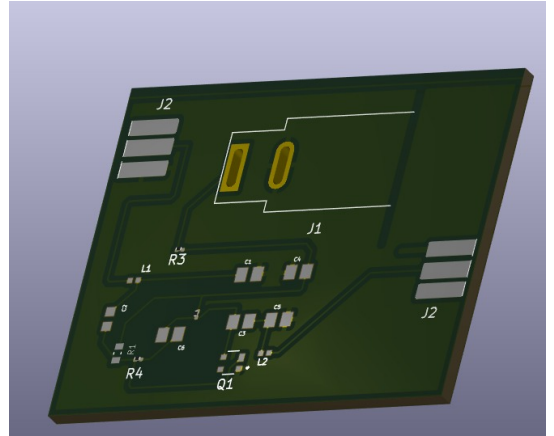


Figure 6: 3D view of PCB design for LNA

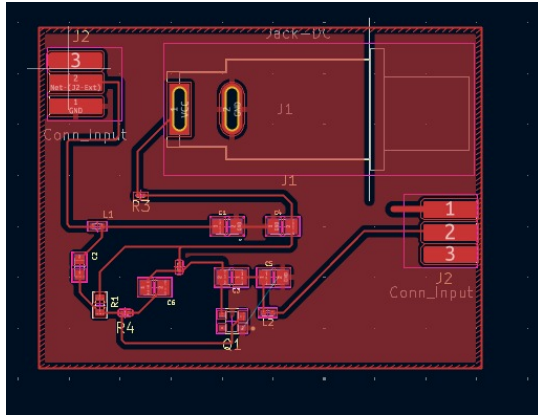


Figure 7: Pathing view of PCB design for LNA

The PCB consists of two SMA connections for signal input and output. A port for DC power supply and the circuitry arranged for proper polarization and coupling of the transistor. The final device implementation is shown below.

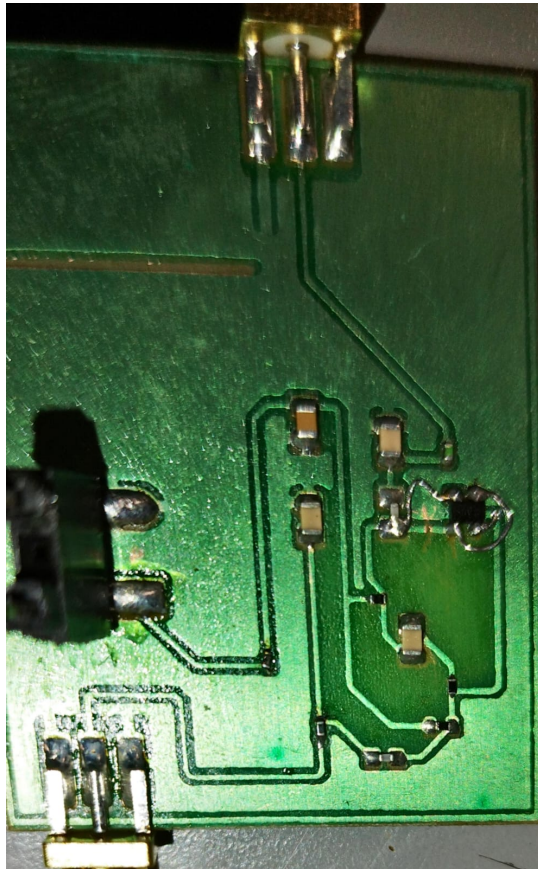


Figure 8: Physical view of PCB design for LNA

B. Results and analysis of the implementation

1) *Spectrum analyzer*: The measurement was taken under the following conditions:

- **Center Frequency**: 100.49 MHz
- **RBW**: 1.000 MHz
- **VBW**: 1.000 MHz
- **Span**: 167.97 MHz
- **Reference Level**: -41.00 dBm
- **Input Attenuation**: 10 dB
- **Peak Amplitude (Marker)**: -50.91 dBm at 100.49 MHz

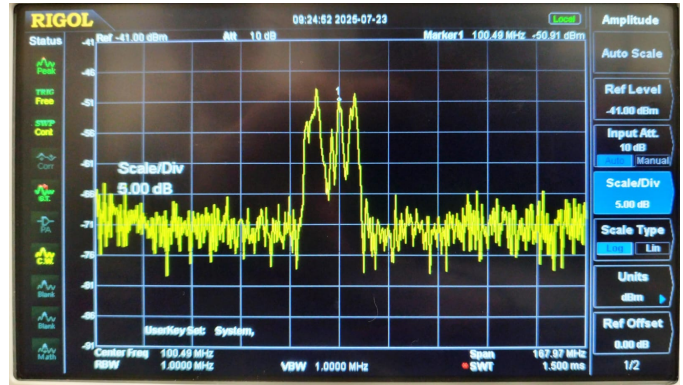


Figure 9: Spectrum analyzer measurement at the output of the LNA.

The signal peak observed at 100.49 MHz corresponds to an FM broadcast carrier. Based on the measured output level of -50.91 dBm, and assuming a typical input signal level from an antenna of about -70 dBm, the estimated gain is:

$$G \approx -50.91 \text{ dBm} - (-70 \text{ dBm}) = 19.09 \text{ dB}$$

This value aligns with the typical gain of the BFP460-based amplifier, confirming proper functionality.

The noise floor is observed around -70 dBm, showing that the LNA introduces minimal additional noise and operates with a low noise figure, consistent with the expected performance of the BFP460 transistor under proper biasing.

The spectrum is clean, with no significant harmonic or spurious components near the main carrier, indicating good linearity and stability of the amplifier. The output shape also reflects the modulated nature of the FM signal, broadened slightly due to the resolution bandwidth.

These results confirm that the LNA is functioning as expected, providing adequate amplification in the FM band while maintaining low noise and good spectral purity.

2) *S₂₁ Gain Measurement of the LNA*: Figure ?? shows the measured S₂₁ gain response of the LNA using a Vector Network Analyzer (VNA), sweeping from 40 MHz to 300 MHz. This parameter indicates the forward transmission coefficient in dB and represents the small-signal gain of the amplifier.

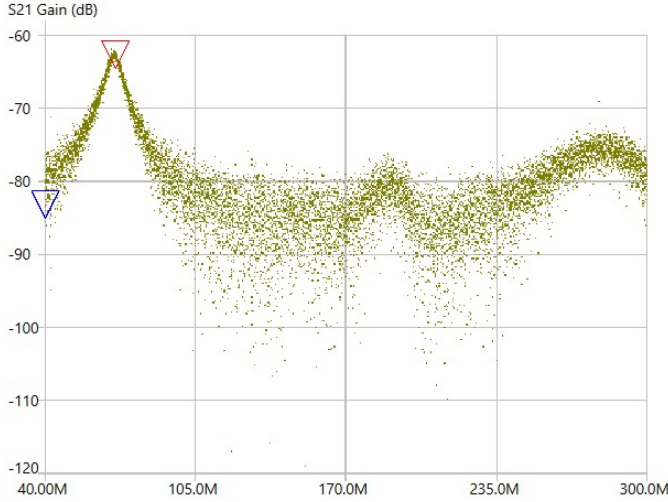


Figure 10: Measured S_{21} gain of the LNA across frequency range from 40 MHz to 300 MHz.

Key observations from the plot are:

- **Peak Gain:** The maximum gain occurs at approximately 70 MHz, with a measured magnitude close to -60 dB (note: this value includes cable and system losses, and is often offset in uncalibrated measurements).
- **Bandwidth:** The gain response shows a narrowband peak near the FM radio band (88–108 MHz), confirming that the amplifier is almost properly tuned for this range.
- **Out-of-Band Rejection:** The gain drops significantly beyond the target band, especially above 150 MHz and below 50 MHz, which indicates good selectivity and helps in rejecting unwanted signals.
- **Gain Shape:** The asymmetric shape of the gain peak is consistent with a narrowband LNA based on a tuned input network (matching through L1 and C1), further supporting the design intention.

This S_{21} profile validates that the LNA is effectively matched and biased to maximize gain in the FM band while suppressing out-of-band signals. The shape and amplitude of the gain curve are consistent with the simulation and theoretical expectations for the BFP460 transistor operated at 2.6 V and 4.6 mA.

V. CONCLUSIONS

The implementation of the low-noise amplifier (LNA) based on the BFP460 transistor at 2.6 V and 4.6 mA demonstrated a functioning amplification stage for the FM band. However, while the circuit generally behaved as expected, several aspects deserve a more critical evaluation:

- **Biasing and Operation Point:** The selection of $V_{CC} = 2.6$ V and $I_{CC} = 4.6$ mA was based on the manufacturer's application note, ensuring operation in the region of optimal gain and low noise. While sufficient to activate

the transistor, further bias optimization may be required for enhanced performance under different environmental conditions.

- **Gain Performance:** Measurements using a spectrum analyzer showed an output signal of approximately -50.91 dBm, suggesting a gain of around 19 dB. This gain is within expectations.
- **Frequency Selectivity and Stability:** The measured S_{21} parameter from the VNA indicates a gain peak near 70 MHz, which deviates from the expected FM center frequency around 100 MHz. This suggests possible inaccuracies in the matching network or resonant components. Additionally, this peak does not align with the main amplification observed on the spectrum analyzer, indicating a discrepancy between the two measurement methods. This mismatch could stem from poor calibration, connector losses, or malfunctioning of the VNA system.
- **Noise and Linearity:** While the noise floor was below -70 dBm, suggesting a relatively quiet amplifier, some asymmetries and distortions were observed in the output spectrum. These could point to moderate non-linearity or insufficient filtering in the output stage.
- **Overall Assessment:** The LNA performs adequately for FM band applications in terms of gain and power consumption. However, the observed shift in frequency response and inconsistency between different measurement instruments point to the need for improved test setup and further refinement of the design, particularly in terms of matching and tuning for frequency stability.

REFERENCES

- [1] Infineon Technologies AG, *BFP460 Wideband Low Noise Transistor Datasheet*, Rev. 2.2, Nov. 2018. [Online]. Available: <https://www.infineon.com/part/BFP460#!?fileId=8ac78c8c7e7124d1017f0227976a6c9d>
- [2] Infineon Technologies AG, *Design Guide for Low Noise Transistor in FM Radio Front-End*, Application Note, Jan. 2004. [Online]. Available: <https://www.infineon.com/assets/row/public/documents/24/42/infineon-design-guide-for-low-noise-tr-in-fm-radio-fe-applicationnotes-en.pdf>
- [3] R. E. Collin, *Fundamentals of Microwave and RF Design*. Raleigh, NC: SciTech Publishing, 2001.